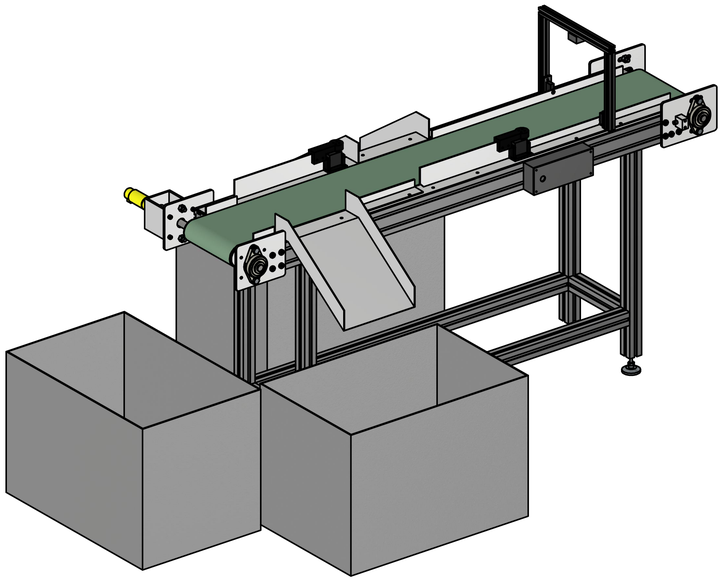
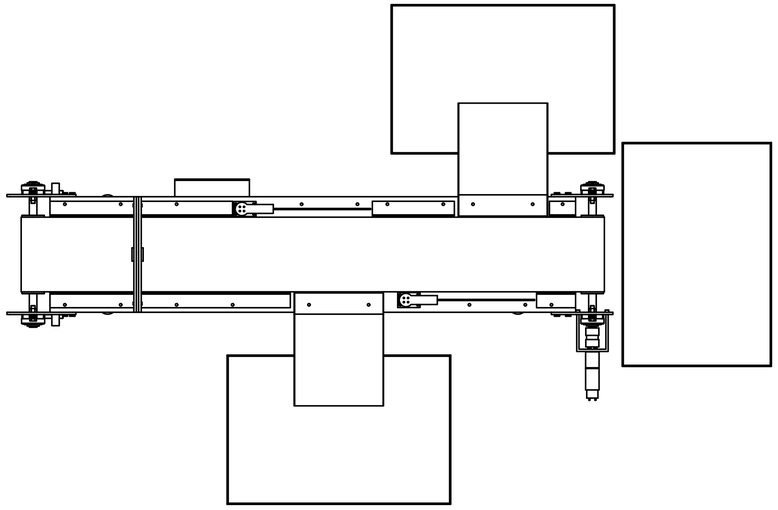
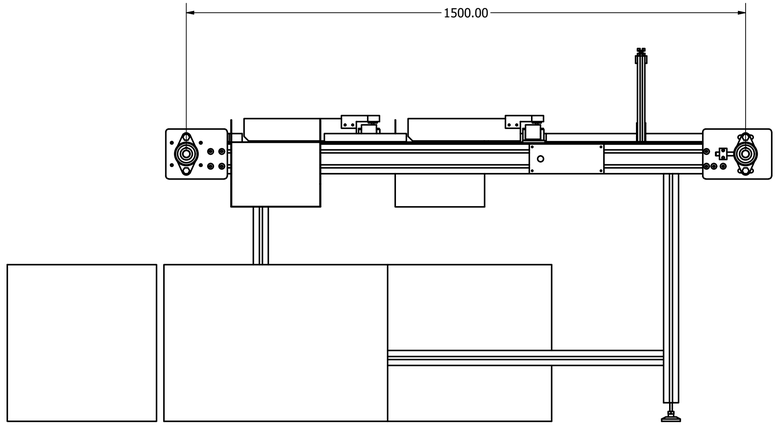
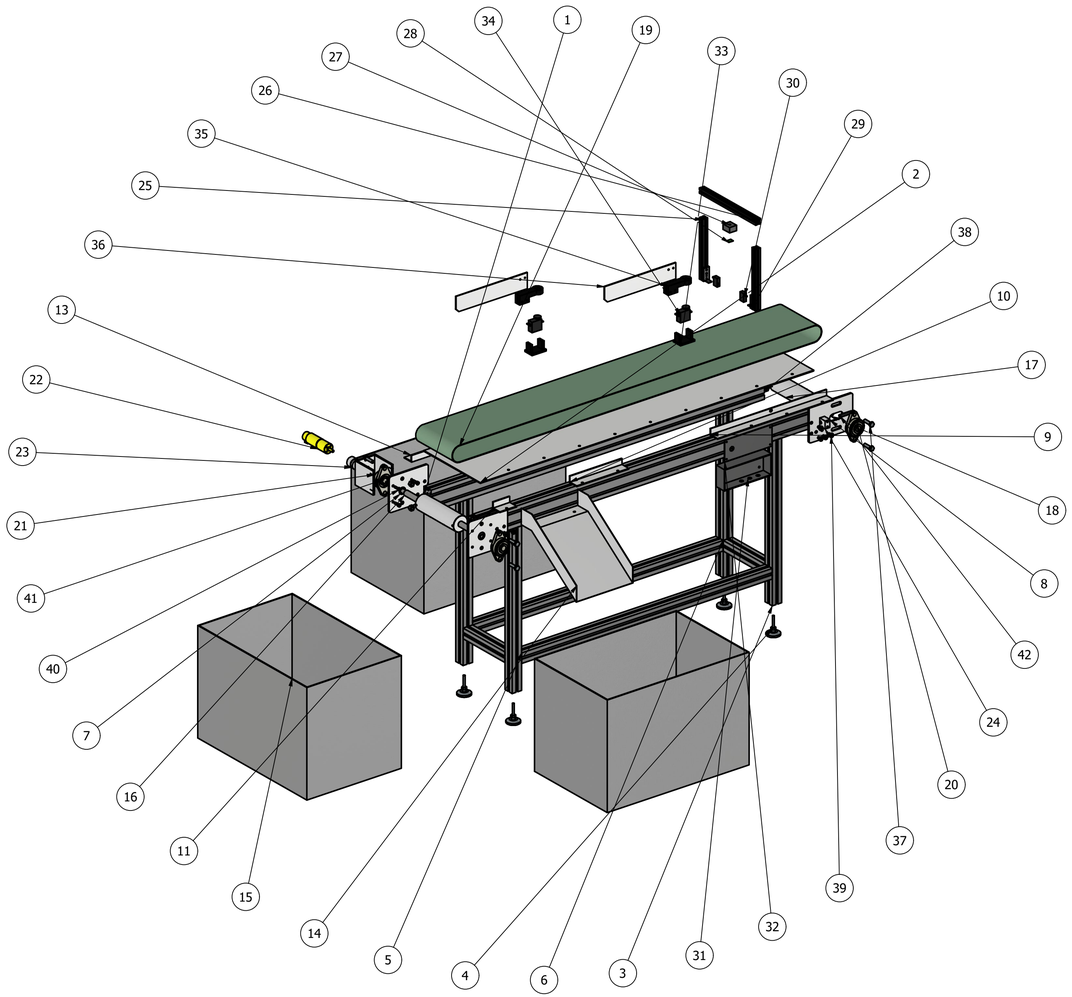
CONVEYORSORTER - AUTOMATED BELT CONVEYOR SORTER, 3 BINS (DESIGNED AND SIMULATED, NOT BUILT)
BELT 200 X 2 PVC, 3195 LONG, 0.25 M/S ON 60 MM CROWNED PULLEYS AT 1500 CENTRES; TAKE-UP +/-15
DRIVE GOBILDA 5203 50.9:1 12 V DIRECT VIA JAW COUPLING: 77 RPM, 0.95 NM NORMAL, PWM 80 %
ITEMS 50-180 L X <=150 W X 30-150 H, <= 1.5 KG; SORT BY L/H (THROUGH-BEAM + VL53L1X TOF AT X 280)
DIVERTERS 2 X SAVOX SB-2290SG, PADDLES 40 DEG; MIN GAP 360 MM -> 1858 ITEMS/H (SIMULATED)
ANCHOR ALL 4 FEET TO THE FLOOR (M8): UNANCHORED THE FRAME TIPS AT 54 N SIDE LOAD AT BELT HEIGHT
BELT TOP 749 ABOVE THE FLOOR; FRAME 1410 RAILS, LEGS AT X 200 / 1300
ASSEMBLY PARAMETERS: TakeUp (TAIL SLOT, X), Divert1 / Divert2 (PADDLE ANGLES)
SHEET 2: EXPLODED VIEW (ConveyorSorter_Exploded.iam) + PARTS LIST







DRAWN Jay Patel	10/28/2026		
CHECKED		TITLE	
QA		ConveyorSorter - general assembly	
MFG			
APPROVED		SIZE D	DWG NO CS-DWG-001 REV
		SCALE 1 / 8	SHEET 1 OF 2



PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
22	1	CS-221	goBILDA 5203 Yellow Jacket planetary gearmotor 50.9:1, 12 V, 117 rpm, 6.71 Nm stall, 9.2 A stall, encoder 1425.1 PPR (envelope; face pattern assumed)
23	1	CS-222	Jaw coupling L-070 class, NBR spider, bores 15 and 8 mm with set screws, 4.9 Nm rated
24	2	CS-206	Take-up block 20 x 30 x 30, 6061-T6, M10 jack-screw thread, 2 x 6.6 to the tail plate (qty 2)
25	2	CS-313	Sensor gantry post, 20x20 slot 6 x 230
26	1	CS-312	Sensor gantry beam, 20x20 slot 6 x 310
27	1	CS-331	ToF sensor head, 3D printed PETG, 30 x 36 x 20, pocket for the VL53L1X carrier, M5 into the gantry beam
28	1	CS-332	VL53L1X time-of-flight distance sensor carrier (Pololu 3415 class), looks down: item height = 207.8 mm - range
29	2	CS-334	Through-beam sensor bracket, 3D printed PETG, 24 x 27 x 48, M5 into the gantry post (qty 2)
30	2	CS-333	Through-beam photoelectric sensor (Omron E3Z-T61 class), 12-24 V PNP, 1 ms response; emitter (+Y) / receiver (-Y)
31	1	CS-341	Control box 200 x 80 x 44, 3D printed PETG, holds Arduino Mega + MD10C driver + buck converter, 3 x M12 glands, M5 to the rail
32	1	CS-342	Control box lid 200 x 80 x 3, 3D printed PETG, 16 mm start/stop button hole
33	2	CS-321	Servo mount, 3D printed PA6-CF (Bambu), base 60 x 40 x 6 + 2 towers carrying the servo ears, 4 x M4 heat-set inserts
34	2	CS-311	Savox SB-2290SG HV brushless servo, 50 kg cm / 0.13 s per 60 deg at 7.4 V, 25T spline, with aluminium disc horn (envelope)
35	2	CS-322	Diverter hub, 3D printed PA6-CF: 36 boss on the servo horn (4 x M3), 84 mm arm, 3.2 slot clamping the paddle with 2 x M4
36	2	CS-323	Diverter paddle 301 x 60 x 3, 6061-T6 laser cut, PTFE tape on the item face, 2 x 4.5 clamp holes
37	8	CS-511	Hex bolt ISO 4017 M10 x 35, 8.8 zinc (bearing flange to end plate)
38	8	CS-513	Prevailing-torque nut ISO 10511 M10, class 8, zinc
39	16	CS-514	Button head screw ISO 7380 M8 x 12 + slot-8 T-nut (end plate to rail)
40	4	CS-515	Socket head cap screw ISO 4762 M6 x 20, 10.9 (motor mount to head plate)
41	4	CS-516	Socket head cap screw ISO 4762 M4 x 10, 10.9 (gearmotor to motor mount)
42	2	CS-512	Set screw ISO 4026 M10 x 50, flat point, nylon patch (take-up jack screw, headless so it clears the plate bolts over the full travel)

PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	2	CS-111	Side rail, aluminium extrusion 40x80 slot 8 x 1410, 6063-T6, both ends tapped M8
2	1	CS-121	Slider bed 1410 x 310 x 3, 5052-H32, laser cut, 18 x 6.6 countersunk M6 into the rail slots; belt runs on it
3	4	CS-112	Leg, aluminium extrusion 40x40 slot 8 x 614, bottom tapped M8 for the levelling foot
4	4	CS-115	Levelling foot M8, 50 mm nylon pad, zinc-plated stud (into the leg core thread)
5	2	CS-113	Lower longitudinal brace, 40x40 slot 8 x 1060
6	2	CS-114	Lower cross brace, 40x40 slot 8 x 230
7	2	CS-131	Head end plate 165 x 134 x 6, 6061-T6: UCFL204 seat (26 + 2 x 11 on 90), 4 x 9 to the rail, 4 x 6.6 motor-mount (qty 2, L/R identical)
8	2	CS-132	Tail end plate 175 x 134 x 6, 6061-T6: take-up slots +-15 mm for the UCFL204, 4 x 9 to the rail (qty 2, L/R identical)
9	1	CS-141	Side guide L1, unequal angle 40 x 25 x 3, 6061-T6, length 490, inner face 103 from the belt centre, 10 mm through-beam window
10	1	CS-142	Side guide L2, unequal angle 40 x 25 x 3, 6061-T6, length 220, inner face 103 from the belt centre
11	1	CS-143	Side guide L3, unequal angle 40 x 25 x 3, 6061-T6, length 70, inner face 103 from the belt centre
12	1	CS-144	Side guide R1, unequal angle 40 x 25 x 3, 6061-T6, length 645, inner face 103 from the belt centre, 10 mm through-beam window
13	1	CS-145	Side guide R2, unequal angle 40 x 25 x 3, 6061-T6, length 105, inner face 103 from the belt centre
14	2	CS-151	Exit chute 240 wide, 1.5 mm 5052-H32, lip 57 + 300 at 35 deg, 60 mm side walls (qty 2, L/R identical)
15	3	CS-161	Collection tote 600 x 400 x 420, polypropylene, stackable (envelope)
16	1	CS-202	Drive shaft 20 h7 x 411, 1045 CD, 15 h7 coupling end with 1 mm flat
17	2	CS-201	Crowned pulley 60 x 210 face (trapezoidal crown 0.2 mm/side), 6061-T6 tube 60 x 5 with integral hubs, 20 H7 bore, 2 x M6 set screws (qty 2, head + tail)
18	1	CS-203	Tail shaft 20 h7 x 390, 1045 CD
19	1	CS-204	Conveyor belt 200 x 2 mm, 2-ply PVC, endless (finger splice), pitch length 3195 mm
20	4	CS-211	Flanged bearing unit UCFL204 (2-bolt, 20 mm, set-screw insert UC204), Cr 12.8 kN, C0r 6.65 kN (NSK catalogue)
21	1	CS-205	Motor mount, 6061-T6 machined U-bracket 84 x 99 x 90: 6 mm front plate (16 pilot, 4 x 4.5 on 25 PCD) + 2 legs tapped M6 into the head plate

DRAWN	Jay Patel	10/28/2026	
CHECKED			
QA			TITLE: ConveyorSorter - general assembly
MFG			
APPROVED			
			SIZE: D; DWG NO: CS-DWG-001; REV:
			SCALE: 1 / 10; SHEET 2 OF 2